INCH-POUND

MIL-PRF-19500/428D
30 August 2000
SUPERSEDING
MIL-PRF-19500/428C
2 July 1999

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTOR, N-CHANNEL SILICON,
TYPE 2N4416A
JAN, JANTX, JANTXV AND JANS

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, junction, silicon field-effect transistors. Three levels of product assurance are provided for the device type as specified in MIL-PRF-19500. (The JANS level is inactive for new designs.)

1.2 Physical dimensions. See figure 1 (TO-72).

1.3 Maximum ratings.

Types	P_T (1) T_A = 25°C	V_{DG} and V_{DS}	V_{GS}	I_G	T_{STG} and T_{OP}
	mW	V dc	V dc	mA dc	°C
2N4416A	300	35	-35	10	-65 to +200

(1) Derate linearly, 1.7 mW/°C for T_A = 25°C.

1.4 Primary electrical characteristics at T_A = 25°C.

Limit	I_{DSS} 1/ V_{DS} = 15 V dc V_{GS} = 0	$V_{GS(off)}$ V_{DS} = 15 V dc I_D = -1.0 nA dc	NF V_{DS} = 15 V dc I_D = 5 mA dc R_G = 1 k ohms f = 100 MHz
	mA dc	V dc	dB
Minimum	5.0	-2.5	
Maximum	15.0	-6.0	2

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAC,Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4 Primary electrical characteristics at T_A = 25°C - Continued.

Limit	$\|y_{fs}\|$ 2/ V_{DS} = 15 V dc V_{GS} = 0 f = 1 kHz	$\|y_{os}\|$ 2/ V_{DS} = 15 V dc V_{GS} = 0 f = 1 kHz	C_{oss} V_{DS} = 15 V dc V_{GS} = 0 f = 1 MHz	C_{iss} V_{DS} = 15 V dc V_{GS} = 0 f = 1 MHz	C_{rss} V_{DS} = 15 V dc V_{GS} = 0 f = 1 MHz
	ms	µs	pF	pF	pF
Minimum	4.5				
Maximum	7.5	50	2	4	0.8

2/ Pulsed width = 100 ms; duty cycle = 10 percent.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

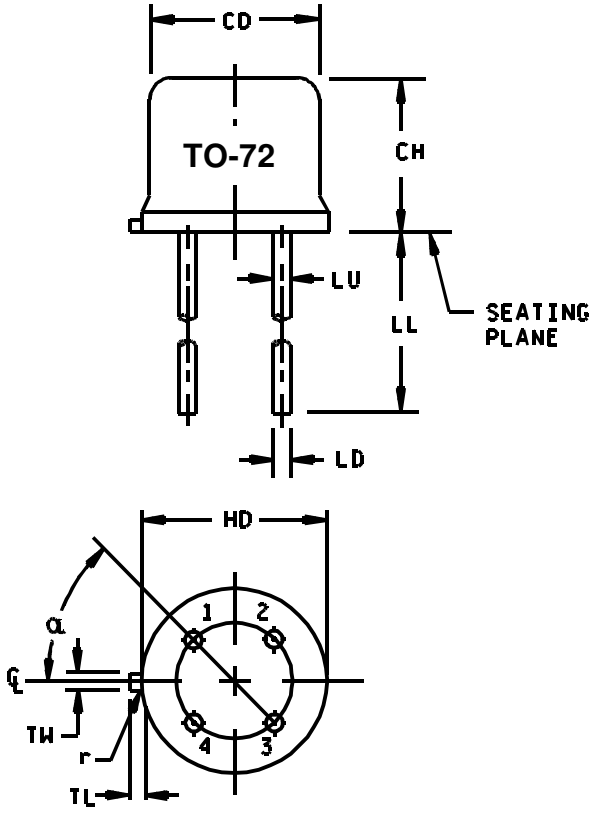
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the, Document Automation and Production Services (DAPS), 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
HD	.209	.230	5.31	5.84	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
LL	.500	.750	12.70	19.05	
LD		.021		0.53	2, 6
LU	.016	.019	0.41	0.48	3, 6
TL	.028	.048	0.71	1.22	5
TW	.036	.046	0.91	1.17	



NOTES:
1. Metric equivalents are given for general information only.
2. Measured in the zone beyond .250 (6.35 mm) from the seating plane.
3. Measured in the zone .050 (1.27 mm) and .250 (6.35 mm) from the seating plane.
4. When measured in a gauging plane .054 +.001, - .000 (1.37 +0.03, -0.00 mm) below the seating plane of the transistor, maximum diameter leads shall be within .007 (0.18 mm) of their true location relative to a maximum width tab. Smaller diameter leads shall fall within the outline of the maximum diameter lead tolerance.
5. Measured from the maximum diameter of the actual device.
6. All four leads.
7. Lead 1 = source, lead 2 = drain, lead 3 = gate, lead 4 = case.

FIGURE 1. Physical dimensions.

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3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-19500.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{G'}$ - - - - - - - - - - - - - Transformed equivalent source resistance.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1.

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be group A as specified herein.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9 and 10	Not applicable	Not applicable
11	I_{GSS1}, I_{DSS}, and $\|y_{fs}\|_2$	I_{GSS1}, I_{DSS}, and $\|y_{fs}\|_2$
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{DSS} = 10 percent of initial value. $\Delta\|y_{fs}\|_2$ = 20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{DSS} = 10 percent of initial value. $\Delta\|y_{fs}\|_2$ = 20 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1039 of MIL-STD-750, condition A; T_A = +175°C; V_{GS} = -24 V dc; V_{DS} = 0 V dc.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in tables VIa (JANS) and VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500, 4.4.2.1, and 4.4.2.2 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	2037	Test condition A.
B4	1037	P_T = 300 mW at T_A = +25°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced air cooling shall be permitted.
B5	2037	(Al-Au die interconnects only) is test condition A.
B5	1027	V_{DS} = 15 V dc; I_D = 20 mA dc at T_A = 100°C or adjusted as required by the chosen T_A to achieve an average lot T_J = +275°C.

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4.4.2.2 Group B inspection, table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	V_{GS} = -24 V dc; V_{DS} = 0; T_A = +175°C ±3°C. No heat sink or forced-air cooling on the devices shall be permitted.
B4	2037	Test condition A.
B6	1032	T_A = +200°C.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C6	1026	V_{GS} = -24 V dc; V_{DS} = 0; T_A = +175°C ±3°C. No heat sink or forced-air cooling on the devices shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Disposition of case lead during electrical measurements. All electrical measurements and operating life test shall be performed with the case lead connected to the source.

4.5.3 Small-signal common-source short-circuit input, output, or forward transfer conductance and susceptance. These tests shall be conducted with a General Radio transfer function and admittance bridge model 1607A (or suitable equivalent) in accordance with the portion of its accompanying handbook which is applicable to this measurement. A Hewlett-Packard generator model 608D and Nems-Clarke receiver model 1502A (or suitable equivalents) shall be used with the transfer function and admittance bridge.

4.5.4 Spot noise figure tests. These tests shall be conducted using the equipment and circuit shown on figures 2 and 3, or suitable equivalent.

4.5.5 Small-signal common-source insertion power gain. This test shall be conducted using the circuit shown on figure 3, or suitable equivalent.

4.5.6 Small-signal common-source short-circuit output capacitance. This test shall be conducted using the circuit shown on figure 4 or suitable equivalent.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, gate to source	3401	Bias condition C; I_G = 1 µA dc; V_{DS} = 0	$V_{(BR)GSS}$	-35		V dc
Gate reverse current	3411	Bias condition C; V_{GS} = 20 V dc; V_{DS} = 0	I_{GSS1}		-0.1	nA dc
Drain current	3413	Bias condition C, V_{DS} = 15 V dc; pulsed (see 4.5.1)	I_{DSS}	5	15	mA dc
Gate to source voltage	3403	V_{DS} = 15 V dc; I_D = 0.5 mA dc	V_{GS}	-1	-5.5	V dc
Gate to source cutoff voltage	3403	V_{DS} = 15 V dc; I_D = 1.0 nA dc	$V_{GS(off)}$	-2.5	-6.0	V dc
Gate to source forward voltage	3403	V_{DS} = 0 V dc; I_G = 1.0 mA dc	V_{GSF}		1	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Gate reverse current	3411	Bias condition C; V_{GS} = 20 V dc;	I_{GSS2}		-0.1	µA dc
Low-temperature operation:		T_A = -55°C				
Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz	$\|y_{fs}\|_1$		11.25	mS
Subgroup 4						
Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz; pulse width = 100 ms; duty cycle = 10 percent	$\|y_{fs}\|_2$	4.5	7.5	mS

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Small-signal, common-source short-circuit input capacitance	3431	V_{DS} = 15 V dc; V_{GS} = 0 V dc; 100 kHz ≤ f ≤ 1 MHz	C_{iss}		4	pF
Small-signal, common-source short-circuit reverse transfer capacitance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; 100 kHz ≤ f ≤ 1 MHz	C_{rss}		0.8	pF
Small-signal, common-source short-circuit output capacitance		V_{DS} = 15 V dc; V_{GS} = 0 V dc; 100 kHz ≤ f ≤ 1 MHz (see 4.5.6)	C_{oss}		2	pF
Subgroup 5						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C delta measurements. 1/ 2/ 3/

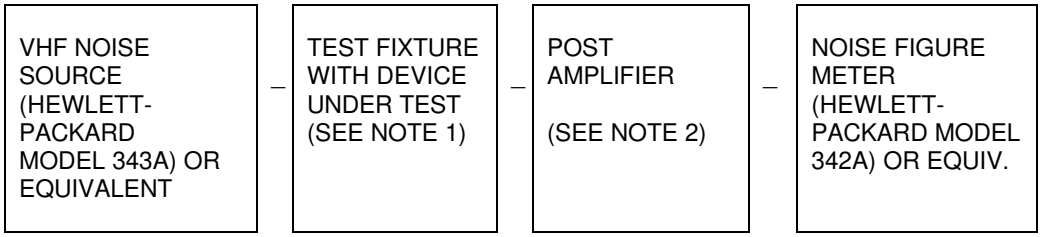
Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Magnitude of small-signal, common-source short-circuit forward transfer admittance	3455	V_{DS} = 15 V dc; V_{GS} = 0 V dc; f = 1 kHz; t_p = 100 mS;, duty cycle = 10 percent	$\Delta\|y_{fs}\|_2$	20 percent change from initial value		
2.	Drain current	3413	Bias condition C; V_{DS} = 15 V dc, pulsed (see 4.5.1)	ΔI_{DSS}	10 percent change from initial value.		

1/ The delta measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 4, see table II herein, step 1.
   b. Subgroup 5, see table II herein, steps 1 and 2.

2/ The delta measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows: Subgroup 3 and 6, see table II herein, step 1.

3/ The delta measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 6, see table II herein, steps 1 and 2; for JANS level and step 1 for JAN, JANTX, and JANTXV levels.

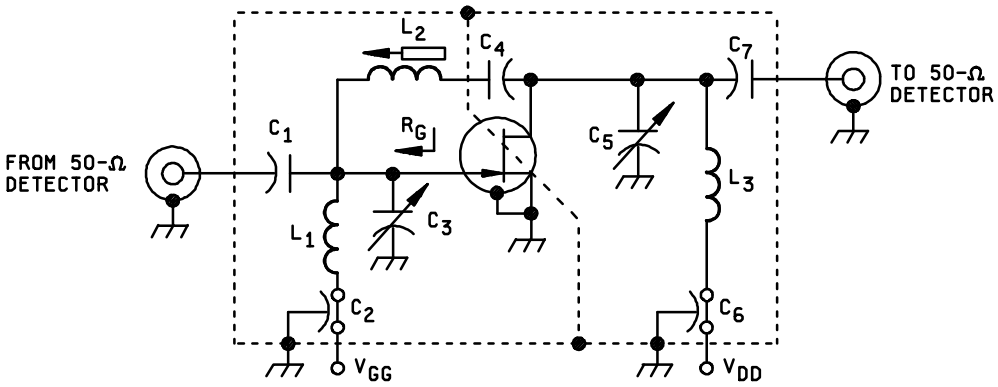
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NOTES:
1. The test fixture and procedure shall be in accordance with figure 3.
2. The post amplifier shall contain suitable biasing circuits and shall be constructed utilizing very high frequency design technique. It shall be designed for the test frequency.

FIGURE 2. Block diagram for spot noise figure tests.

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Circuit component information (see note)			
	Capacitors	Inductors	
	100 MHz		100 MHz
C_1 C_2	7 pF 0.0015 µF	L_1	0.14 µH, 3.5 T, no. 18 enameled copper wire, 3/8" I.D., 1/4" long
C_3 C_4 C_5	1-12 pF 1,000 pF 1-12 pF	L_2	3 µH, 17 T, no. 28 enameled copper wire, close wound, 9/32" I.D., powered iron slug
C_6 C_7	0.0015 µF 3 pF	L_3	0.25 µH, 4.5 T, no. 18 enameled copper wire, 3/8" I.D., 5/16" long

NOTE: Transformed equivalent source resistance is 1,000Ω at 100 MHz for 100 MHz amplifier.

FIGURE 3. Neutralized small-signal common-source insertion power gain and common-source spot noise figure test circuit.

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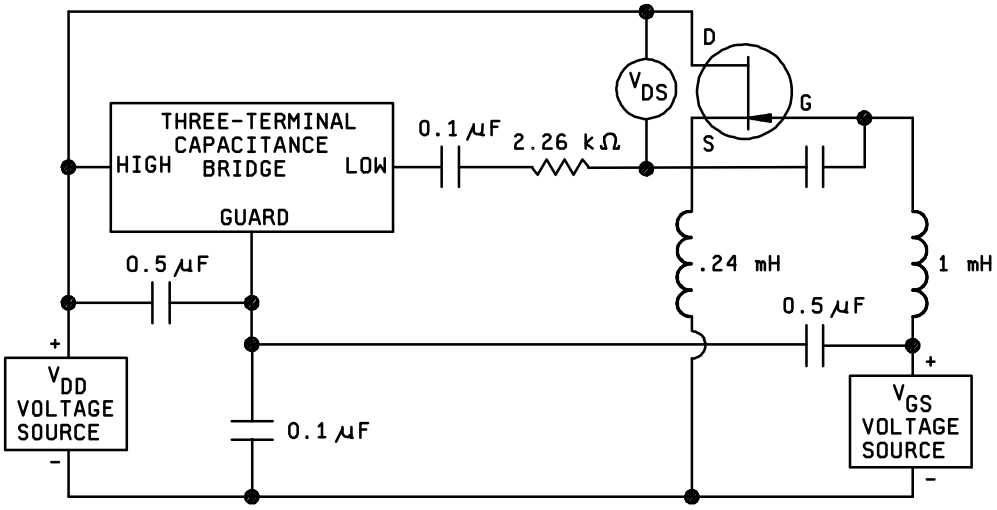


FIGURE 4. Small-signal common-source short-circuit output capacitance test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements (see 5.1).

d. Lead finish as specified (see 3.4.1).

e. Product assurance level and type designation.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC-VQE, P.O. Box 3990, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - MC
Air Force - 19

Preparing activity:
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/428D	2. DOCUMENT DATE 30 August 2000

3. **DOCUMENT TITLE**
SEMICONDUCTOR DEVICE, FIELD-EFFECT TRANSISTOR, N-CHANNEL SILICON, TYPE 2N4416A, JAN, JANTX, JANTXV AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) COMMERCIAL DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510	DSN 850-0510	FAX 614-692-6939	EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAC P.O. Box 3990 Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888			

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